

Spencer Bowden

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EDUCATION

VANDERBILT UNIVERSITY, PEABODY COLLEGE (NASHVILLE, TN)

2022-2026

Bachelor's of Science: Cognitive Studies, Computer Science, Minor in Music (Piano Performance Concentration)

GPA: 3.88, Dean's List (all semesters attended)

Relevant Coursework: Data Structures, Algorithms, Design Patterns, Game Engine Arch., Discrete Math, Neural Networks, DBMS
Linear Algebra, Differential Equations, Probability and Statistics, Web Development/Systems, UI/UX

SKILLS

Languages: C++, JavaScript, Python, Java, Rust

Full-Stack: Front-End : React, JavaScript, TypeScript | *Back-End:* Django, Express.js/Node.js, Firebase | *Databases:* SQL (Postgres), MongoDB

Game Design: C++, Unreal Engine 5, Godot Engine, C#, Rust

DevOps/Tools: Git/GitHub, AWS, Linux/Ubuntu, Docker, CI/CD, Figma

PROJECTS

FAUNOLOGY— *React, Vite, Phaser.js, Firebase*

- Led full-stack integration on a collaborative team, owning end-to-end application architecture while teammates owned game logic and art.
- Developed a browser-based cozy tower defense game integrating Phaser.js within a React/Vite architecture, bridging game loop logic with in-game/out-of-game component-based UI
- Implemented Firebase Auth, Firestore-backed save state persistence and Firebase Hosting; shipped to 25+ active users.

SPENCER SOLITAIRE SAGA — *React, Node.js, MongoDB, AWS (EC2, Route 53)*

- Built a full-stack solitaire app with a React front-end and Node.js REST API handling game logic, move validation, and JWT-based authentication.
- Persisted card states, move history, game state, and user profile data in MongoDB; designed schema to support full game session reloads.
- Deployed to AWS EC2 with HTTPS configured through AWS Certificate Manager and custom domain routing via Route 53

PROFESSIONAL EXPERIENCE

TRITON SENSORS (REMOTE, HARRISBURG, PA)

2026

Full-Stack Summer Development Intern - React/TypeScript + Django REST API

- Built full-stack features on Triton's cloud dashboard (React/TypeScript + Django REST API) for a smart-sensor safety platform deployed in schools.
- Shipped multiple end-to-end projects, including a 12/24-hr time-format setting for clients, a permission-gated audit diff viewer for account changes, a scheduled alert report generator sent to emails, and SVG-based navigation loading animations.

VANDERBILT UNIVERSITY, PEABODY COLLEGE (NASHVILLE, TN)

2025-2026

Undergraduate Research Assistant (Computational Cognitive Development Lab — PI: Dr. Deon Benton; VU Summer Research Program 2025)

- Crafted my own research regarding contrastive learning found in children's cognition through on-site psych studies and neural networks
- Constructed neural networks in Python and Tcl/Tk that model learning to further understand computational nuances behind cognition

VANDERBILT UNIVERSITY, SCHOOL OF ENGINEERING (NASHVILLE, TN)

2024-2026

Undergraduate Teaching Assistant (CS1101: Programming and Problem Solving ; CS2201: Program Design and Data Structures)

- Hosted weekly office hours for student tutoring on Computer Science course material and general Q&A; Assessed exams/Java PAs

LEADERSHIP EXPERIENCE

VANDERBILT OFF-BROADWAY (VOB) | VANDERBILT UNIVERSITY (NASHVILLE, TN)

President

2025-2026

- Led 12 executive board members to supervise the daily operations and successful musical productions of the 100+ member organization
- Oversaw officer selections, interviews, and artistic team training
- Maintained VOB's constitution and campus presence by coordinating performances, ensuring annual registration

Music Director (Chicago: 2024 | Natasha, Pierre, and the Great Comet of 1812: 2024-2025 | Into the Woods: 2025-2026 | Falsettos: 2026)

- Led vocal rehearsals/office hours with show casts, focusing on intermediate musical theatre vocal technique and good vocal health
- Arranged and led pit orchestra rehearsals with 10+ instrumentalists, as well as played and conducted during performances
- Collaborated with artistic team to ensure the show's musical facets fit the director's vision, composer's intentions, and production's goals